<table>
<tr><th colspan="3" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>2</td><td>(a)</td><td></td><td>Vector: Magnitude (size) and direction
Scalar: Magnitude (size) only
Minimum acceptable response: a vector has direction [1]
Relevant example of each e.g. independent mark [1]
<table><tr><th>Vectors</th><th>Scalars</th></tr><tr><td>Displacement
Velocity
Acceleration
Force</td><td>Speed
Time
Distance
Pressure
Temperature</td></tr></table></td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td>From graph speed = $\frac{2\,000}{100}$ or equiv [e.g. $\frac{1000}{50}$] (or 20 m s^{-1}) seen [1]
Convincing conversion to km h^{-1} (i.e. $\times\ \frac{3\,600}{1000}$) or 72 [km h^{-1}] seen or convert 70 km h^{-1} into 19.4 m s^{-1} [1]</td><td>1</td><td>1</td><td></td><td>2</td><td>2</td><td></td></tr>
</table>

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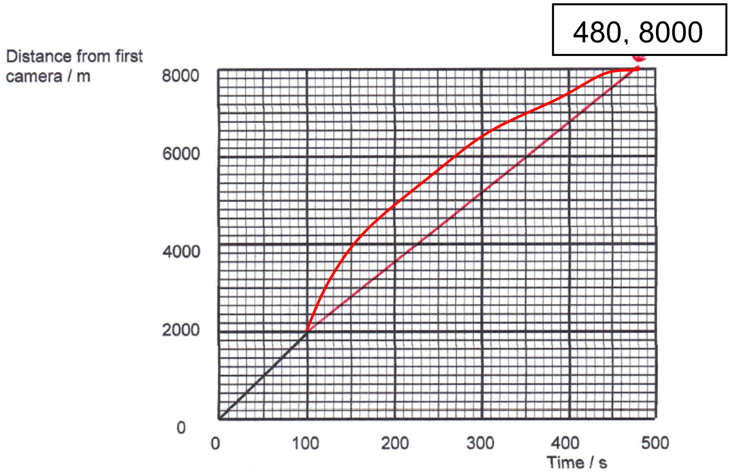
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Time to complete distance at mean speed of 60 km h^{-1} = $\frac{8}{60}$ [1] (= 0.1333) Conversion t = 480 s [1] Accept rounded values e.g. 17 m s^{-1} gives t = 471 s **Alternative for first 2 marks:** 60 km h^{-1} = $\frac{1000}{60}$ m s^{-1} (1) $t = \frac{8000}{\left(\frac{1000}{60}\right)}$ = 480 s (1) **Alternative for first 2 marks:** 1 km per minute (1) t = 8 minutes or 480 s (1) **3rd mark** - Continuous (though not necessarily straight) line drawn on graph from (100, 2 000) to (480**ecf**, 8 000) tolerance ± small square - two possible examples shown: [1] 		3		3	3	

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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Correct substitution into $v^2 = u^2 + 2ax$ (i.e. 0 = u^2 – (2 × 8 × 85) [1] don't award mark if incorrect signs u = 36.9 [m s^{-1}] **ecf** on incorrect signs [1] No, greater than speed limit [1] must correlate with working **Alternative:** Correct substitution into $v^2 = u^2 + 2ax$ using 30 m s^{-1} to find x. (i.e. 0 = (30)^2 – (2 × 8 × x) [1] don't award mark if incorrect signs x = 56.3 [m] **ecf** on incorrect signs [1] No, greater than speed limit [1] must correlate with working **Alternative:** Using $v = u + at$ to determine t = 3.75 s (1) Using $x = \frac{1}{2}(u+v)t$ i.e. 3.75 × mean velocity (15) = 56.3 [m] (1) No, greater than speed limit [1] must correlate with working			3	3	2	
			Question 2 total	**3**	**4**	**3**	**10**	**7**	**0**

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